Printed Pages: 14 University Roll Number:

B. Tech II Year, III Semester, End Term Exam-2014

AHE-201: Ethics & Values

Time: 2 ½ Hours **Max. Marks: 40**

Note: This paper consists of four passages. The passages are not unseen and are based on course discussion. They help in your analytical problem solving. Carefully read the passages and answer the following questions having the multiple answers. Carefully fill all the information in OMR sheet neatly in the blocks. No partial marking.
Full marks (assigned) will be given for a complete answer that means all correct choices.

Section A

Passage 1 (10X1.5 = 15) (Attempt All Questions)

Justice is to respect according to what one deserves. In other words, if one performs good meritorious acts one must be rewarded and if one does something wrong one must be punished accordingly. It is the argument from moral desert that theorists define justice in a manner a person actually performs and deserves accordingly. However, the moral desert argument should be understood in contrast to the affirmative action theory by which a government usually acts positively in redressing the justice to benefit the underrepresented group in the areas of education, employment and business. There are three prominent views on the justice and its distribution to the common people. Let's first talk about the John Rawls who in his a Theory of Justice advances a thought experiment what he calls "Original Position" is actually a situation in veil of ignorance in which all so called self-interest seeking individuals, who are conscious and psychologically fit actors to take decisions or make choices on the benefits and developments, have no knowledge of their arbitrary facts say race, physical disability, cast, gender and class. Now once veil goes up they have to choose and decide on the better principles of justice by which a fair distribution of

resources, powers and rights can be done. Rawls argues that there can be two principles seem reasonable in such a situation: 1) Equality principle, according to which each individual must be provided equal rights with the most extensive basic liberties compatible with similar liberties for others, and 2) Difference principle what he calls qualified equality principle that maintains that socio-economic inequalities have to arranged in a manner so that they are to be the greatest benefit of the least advantaged members of society. Here it seems that Rawls tries to balance his theory of justice as fairness with the theory of distributive justice or social justice. Before we go to understand what Rawls actually mean by his difference principle let us try o understand three basic arguments of affirmative action theory: 1) Corrective argument; according to which a government should take positive actions to redress the wrongs of past by benefiting some cast, race, gender or group who have not been given the proper resources to develop and compete. 2) Compensatory argument which points out the fact that some advantaged groups of people have to compensate for some least advantaged groups of people who have been deprived of basic resources for development, and 3) Social Representation / Diversity argument that maintains that the different variety of groups who actually make up the whole society should be given proper and proportionate socio-political-economic representation in terms of education, job and business. However, Rawls in his qualified equality principle argues that socio-economic inequalities in terms of government affirmative action policies and provisions must be in favor of least well-off. The second prominent view in contrast to distributive theory is Aristotelian Virtue theory in which he criticizes the Utilitarian view of justice in favor of what he calls Teleological view of justice. Aristotle asks: if you have to distribute best flutes to who would you distribute? Generally, we come with responses like needy, interested or who can use best the flutes will deserve the best flutes. However, surprisingly, Aristotle denies these responses and says the best flutes have to be distributed to the best flute players who have the best excellence or virtues to practice the flute playing. Here, Aristotle offers his teleological explanation to this idea of justice that

society is consists of different kinds of people having different virtues and people must be rewarded according to their best virtues by which they practice those virtues and merits at best to create a best society. Thus, justice in fair distribution has got a telos or purpose to give resources and opportunities to the people so that that can create a good society by practicing their virtues. Aristotelian view really throws light on a serious issue called reservation. A group of people who are suffering from the socio-economic equalities in education, job and business usually criticizes the other benefited groups and argue that there should not be inequalities be maintained by the governments through affirmative or positive actions. They further argue that due to reservation less qualified and skilled people get the powers and positions in governance and they produce poor quality results. But, this view is questionable on the grounds that the people of either group advantaged or non-advantaged, meritorious or less qualified are doing the same thing corruption in the various systems without practicing their skills or virtues. As a result, the so called merit is not serving the purpose to create a better society and most of the people those who are getting opportunity and resources are utilizing and exploiting their powers and positions. Nevertheless, the reservation given in the name of cast and other arbitrary facts which are only by chance and not chosen by any individual is really poses a moral problem of forced discrimination. Therefore, the reservation policies must be in Rawlsian terms be based on qualified equality principle serving the least well-off to really improve the socio-economic condition of less advantaged people. The third prominent view offers a criticism of both the Rawlsian and Aristotelian views of justice. The third view advocates individual rights theory and of against active roles of government in fair distribution of resources. Immanuel Kant and Robert Nozick are the two main philosophers who actually defend the theory of rights and liberty of people. Immanuel Kant argues that individuals have rationality to decide their better off and thereby they must be given autonomy to choose their own ideas of good life. We should not lead by any purpose to create a better society; rather a better society will automatically come when the

individuals will be given basic liberties and rights with a sense of autonomy. On the other hand, Robert Nozick rejects the active roles of governments and strong forms of affirmative action theories. He argues that government roles should be limited to the security and protection of people. There should not be any paternalistic legislation, moral legislation, and wealth re-distribution through taxation. It is in this way a government should not intervene in individuals' lives and should not limit their choices relating to education, job, business and life as a whole. The first two views and the third view in contrast get impression of a tussle between the Fit versus freedom. Rawls and Aristotle argue about the fitness of the principles and virtues, while Nozick and Kant defend the rights and freedom of individuals. However, Rawls rejects the Nozick's point that merit is the criteria for just distribution. He says merit is just a moral luck and no one can claim moral credit for his merits. Merit is arbitrary fact that no one creates by his own. No one has ownership of the merit. We only have the legitimate ownership of the merits.

Only reservation is not the resolution and creamy layer must be excluded from the benefit of reservation. And, better executed reservation must not solving the purpose of cast-vote-politics, rather, it should purposefully lessen the socio-economic gaps amongst the people so that they can all compete in the race of development with ,if not an equal start then at least ,a fair start. Reservation in this way rationally be based on socio-economic real facts and conditions, and the socio-economic benefits should be distributed to those who really deserve without considering the arbitrary facts to which cast, class, race and gender they belong. Indeed we cannot really go in past to rectify what have done but taking meaningful lessons from the past it is better to give proper resources to all those who deserve and want to compete and develop. It is good to create a better present than making it again an unjust past for some groups of people in coming future.

1) Which of the following points are right about the moral desert argument?

➢ One should be respected on the grounds one actually deserves.

- One should be rewarded for one's good actions and punished for the bad.
- Justice should be based on one's effort.
- Socio-economic inequality should be for least well-off.

2) "The best flutes must go to the best flute players according to Aristotle". Which of the following statements **cannot** be said to be right according to Aristotle?
- Best flute players require best flutes.
- Best flute players derive pleasure in best flutes.
- Best flute players would use flutes at best.
- Best flute players have virtue to practice flutes at best.

3) What is veil of ignorance?
- It is a thought experiment.
- It is the position in which people are unknown to their arbitrary facts.
- It is the position where we don't know about our history.
- In this people are known to their arbitrary facts.

4) Which of the following statements are **NOT** true about Affirmative Action Theory?
- A government should not take positive actions to redress the wrongs of past.
- Some disadvantaged groups of people have to compensate for some advantaged groups of people.
- The different variety of groups who actually make up the whole society should be given proper and proportionate socio-political-economic representation.
- Socio-economic inequalities should not be based on arbitrary facts.

5) What is Teleological argument of justice?
- Different kinds of people having different virtues create a society.
- People must be respected according to their best virtues.
- Justice has a purpose to create a better society.
- Justice is a distributive notion.

6) Which of the following statements are right about the Difference Principle of John Rawls?
- It is the principle of qualified equality.

- It allows the socio-economic inequalities that are best for least well off.
- It respects both one's efforts and the benefit of disadvantaged people.
- Difference principle only respects merits.

7) Which of the following statements are **NOT** correct about corrective and compensatory arguments?
- A governments should not take any positive actions to correct the past wrongs.
- Those cast, race, and gender groups will be benefited who have been privileged in history.
- Advantaged groups have to compensate for least advantaged group.
- There should not be any discrimination.

8) What does Robert Nozick criticize the affirmative action theory?
- Because a government take positive actions to intervene in individuals lives.
- Because a government re-distribute the wealth in the name of taxation to benefit unprivileged people.
- Because a government limit their choices relating to education, job, business and life as a whole.
- Because a government protect and secure people from external threats.

9) What is criticism of I. Kant on the Aristotelian view of justice?
- Every individual can have a different autonomous purpose.
- People have rationality to decide their better off.
- People must be given autonomy to choose their own ideas of good life.
- We should respect only virtuous people who are fit for society.

10) What are the views of Robert Nozick that John Rawls would **Not** accept?
- Merit is criteria for just distribution.
- Merit is one's ownership.
- We have moral credit to our merits.
- Merit is just our moral luck.

Section B

Passage-2 (Attempt All Questions: 10X1= 10 Marks)

Libertarians maintain the two fundamental individual rights are: 1) Freedom and, 2) Respect. Theories of individual rights make the point clear that each individual has a separate life and different preferences. Individuals' preferences should not be sacrificed for either larger social purposes or the common good. We people not only desire but also value the freedom. By the very human nature we like freedom and do not want to live in the restrictions of laws and rules. Nevertheless, we desire those minimum laws and rules which actually protect ourselves from others. Libertarians define freedom in such a way that freedom for individuals is all or nothing. In other words, they argue individuals must be given complete freedom. However, Libertarians differentiate between individual and collective (social) freedom. According to libertarians, there should not be any kind of limitations on individual freedom. However, there is a condition for collective freedom that one is free to the extent that one must not harm the other. Most interesting feature of the Libertarians' idea of freedom is that one has got complete freedom in a very precise sense that one can, if one wishes so, harm himself or herself. Libertarians advocate for the minimal state theory.

What this theory simply means is that state should not limit the liberty of individuals or a government should not regulate the individuals' choices and actions by propounding certain restrictions and laws. Individuals have got fundamental right to liberty. Government has got only role to play in individuals' lives, and that is to protect them from any kind of external threat like theft, fraud, and corruption. Libertarians object the excessive use of the powers of government in the way it intervene in individuals' lives and coerce them to follow certain laws. There are three such objections by Libertarians against the governments. First, government should not make paternalistic laws. Second, government should not make laws for the morals legislation of individuals' lives. Third, government should not redistribute wealth by putting taxes on individuals' salary. By paternalistic laws we mean such laws which are

actually for the individuals' safety and protection like the helmet. seatbelt driving safety laws, and the pension laws. Libertarians argue that enforcing such laws on an individual a government undermines his or her risk taking ability. Libertarians believe that government should not propose a certain kind of values or principles for individuals' lives. An individual must be given full freedom to choose his or her own patterns and ways to live the life. However, Libertarians put one condition on this that one's moral choices must not affect the others. On the issue of taxation, Libertarians contend the government and say: by putting taxes on individuals' income government takes away the ownership of an individual. The amount we pay as taxes becomes slavery for an individual. What we earn through our labor talent or skill is our self expression. In this way, a self becomes slave and one's labor becomes forced labor. This is the reason governments should not redistribute the wealth by following the mechanism of taxation. However, governments can have a check on individuals' property rights. A government must insure fair dealings amongst the individuals in terms of the initial holding of property, justice in acquisition and transfer of money, and other property rights. Libertarians by limiting the excessive role of government maintain: A government should respect an individual's personal life. The life of an individual must be respected as an end and never as means. Respect an individual's life projects independent of collective common goods. And, by respecting an individual a libertarian simply means: government should not interfere in individuals' personal life for accomplishing the purposes of society at large.

Questions:

1) There should not be paternalistic laws. What we can infer about libertarians from this statement?

- Libertarians object paternalistic laws.
- They maintain that such laws restrict one's freedom.
- Such laws undermine one's risk taking ability.
- Such laws are consistent with one's freedom.

2) What are the reasonable demands of the defenders of Individual Rights?

- Individual choices should not be sacrificed for larger social purposes.
- Everyone has separate life.
- Each individual should be given complete freedom and respect.
- An individual can restrict others rights.

3) Which of the following statements are **wrong** about Libertarians?

- Individual freedom is nothing.
- Liberty is the fundamental right of an individual.
- An individual own himself or herself.
- We can sacrifice our freedom for society.

4) Libertarian theory of Individual Rights is Naturalistic because:

- It's idea of freedom based on human nature.
- It rejects other's rights.
- It maintains that an individual desires freedom and does not like rules and restrictions.
- It argues that we like to be free and respected.

5) Which of the following statements are **False** about Morals legislation?

- Moral legislation is about giving freedom to individuals to practice their life on their choices.
- Limiting certain choices of individuals which disturb socio-cultural values.
- Allowing people to choose their own ways of life.
- It is about some virtues and values that a government wants to be practiced by a society.

6) What is minimal state theory?

- A government should not intervene in one's life.
- A government must control individual life.
- A government has some minimal roles for people.
- A government should only protect individuals from external threats.

7) What are the check and balances of a government that the libertarians allow?
   - A government should have check on just transfer of money.
   - A government should have a check on one's initial holdings.
   - A government should check the acquisition of money.
   - A government should acquire the property of people.

8) What is respect according to Libertarian?
   - An individual should be respected as an end.
   - An individual must not be treated for larger social purposes.
   - We should respect one's life project.
   - Sometimes we can treat an individual as an end.

9) Why is taxation wrong according to Libertarian?
   - It rejects ownership of an individual.
   - It makes one's profession a forced labor.
   - Taxation is right.
   - Taxation is denial of one's complete freedom to his or her salary.

10) Why would the libertarian allow Euthanasia?
   - An individual has complete freedom to oneself.
   - An individual has complete rights to decide about his life.
   - We should respect one's choice if it does not harm the others.
   - We cannot take our life as it is against our individual right.

Section C

Passage: 3 (6X1=6) (Attempt All Questions)

According to moral desert argument one deserves on his or her merit, skills and efforts. In the very ancient time Aristotle made this point clear that justice is giving people what they deserve. But how do we know what people deserve? What goods and opportunities should go to which persons? Aristotle's answer is that we have to consider the telos—the point, the end, or the purpose—of the good in question. Say we have some nice flutes. Who should get them? According to Aristotle, it's not the rich person, since playing flutes has nothing to do with money, nor is it the person who will be made most happy, since making good music is different than being happy. The purpose of a flute is to be played and to

be played well. Therefore, Aristotle thinks, the flutes should go to the best flute players. Aristotle's method is to think about justice by thinking about the purpose of a good, an institution, or even a person. If the purpose of a tennis court is to play tennis, then the best tennis players should get priority. If the purpose of universities is to create knowledge and to foster excellence, then the best students should be admitted and the best scientists and professors should be allowed to work at them. If the purpose of being human is to live a good life, then society should promote the good life by ensuring that citizens have the resources necessary for living a good life and by encouraging them in the pursuits that make for a good life. Is this the right way to think about justice?

Questions:

1) We should respect the virtues. Which of the following points are consistent with this Aristotelian view?
   - The best flutes must go best flute players.
   - We should distribute according to merits.
   - The best flute players morally deserve to get best flute.
   - The best flute must go who can best utilize it.
2) What is teleological view of justice?
   - The purpose to create a just society.
   - It is the purpose of an institution or even a person.
   - It is to create a better society.
   - It is to fulfill need.
3) Which of the following statements **cannot** be said to be the purpose of society according to Aristotle?
   - It is to provide a good placement.
   - It is to give admission needy people.
   - It is to admit best student.
   - It is to allow best scientist and professors to work.
4) Which of the following are **Not** the questions about the Aristotelian view of justice?
   - How do we know what people need?
   - What goods and opportunities should go to which persons?
   - Who can best utilize the resources?

- What is the best utility of a resource in distribution?

5) Which of the following statements are right about justice according to Aristotle?
   - Justice is based on moral desert.
   - It is giving what one deserves.
   - It is respecting humans as an end.
   - It is respecting one's virtues and merits.
6) The purpose of justice is to create a better society. What we can conclude from this statement?
   - Virtuous people should be respected.
   - Justice is about social justice taking care of all irrespective to the merits.
   - Better society can be formed only by best people.
   - Only meritorious people can create better society.

Passage: 4 (6X1.5=9) (Attempt All Questions)

The principle of utility: maximum good for maximum number of people is the theory grounded in human nature. It is the balance of good over bad and happiness over the sufferings. In this moral theory, we make a choice on the basis of maximum utility, and a Utilitarian defines maximum utility in terms of the results or the consequences an action produces. A Utilitarian argues that a right choice/action/decision is something that produces maximum good for maximum number of people. But, they do not care about preferences and choices of an individual. However, a Utilitarian takes care of a society as a whole. Jeremy Bentham, in his work on Utilitarianism, defines that by the very human nature we desire pleasure and avoid pain. And, whatever provides pleasure over pain is preferable as a right choice for Bentham to act on. This is a naturalistic way of defining the rightness of a choice or an action. Although John Stuart Mill follows Bentham and is agreed with him on the most of the points of Utilitarianism, Mill makes a distinction between lower order and higher order pleasure. He says a higher order pleasure is something that involves human capacities of thinking, differentiating, and judgment and lower order pleasure is simply we enjoy pleasure without testing the nature of pleasure on the

grounds of human rational capacities. Here, we can understand that though Mill gives preference to pleasure and happiness but he gives more importance to rational features of human beings in making qualitative choices and differences amongst the pleasures of different kinds. For an example, a drug addict who derives pleasure in the addiction of drugs only minds the quantity of pleasure s/he gets through it. S/he does not rationalize the consequences or the result of drug addiction. However, the reformative and rehabilitative programs running across the whole world against the drug addiction are working on the suggestions of Mill that we should rationalize the consequences or results in terms of quality of pleasures. Once this qualitative difference making capacity is being realized by a drug addict can get rid of addiction and can live qualitatively a better life.

Further, Mill develops an interesting point on the idea of freedom that freedom is something that cannot be assessed on the grounds of utility. In other words, he argues that the criteria of defining the right things to do on the grounds of maximum good of maximum number is fine to the extent it does not affect ones fundamental rights say freedom. But, if a maximum good of maximum number is overriding individual rights then Mill argues he would not go with Utilitarianism. He says; if it comes to chose between the Utilitarianism and Individual Freedom, I would choose individual freedom over the utilitarianism. Let's understand this argument by a prevailing problem of Indian culture namely, Honor killing. The issue of Honor Killing involves a conflict between individual choice and social sanctioning (Social legitimacy). If we analyze this problem on the views of Jeremy Bantam then we would say Honor Killing is justified since it takes care of the preferences of maximum number (Particular society) in order to define the right thing to do. However, Mill would object Honor Killing because here maximum good is overriding individual rights and freedom. In this way, we find Mill more reasonable in selecting the quality of pleasure and happiness and defining utility of that accordingly with the help of

reason. This is what we can say rise of reason for the moral judgment decide what is right and what is wrong.

Questions:

1) Mill would reject the Honor Killing because:
- It is against one's individual freedom.
- Individual choice cannot be sacrifice for maximum good.
- Mill prefers individual rights over maximum good.
- Mill would not mind sacrificing individual choice for maximum good.

2) What is the principle of utility?
- Maximum good for maximum number.
- Utility is balance of happiness over sufferings.
- Utility is balance of good over bad.
- Utility is to respect one's rights.

3) Jeremy Bentham and Mill both would **Not** agree on these following prin
- Humans desire happiness by nature.
- Utilitarianism suits to human nature.
- A qualitatively better pleasure produces more utility than a qualitati inferior pleasure.
- Individual rights are more important than utility.

4) Mill prefers to have higher order pleasure because:
- It involves thinking and judgment.
- It involves differentiation.
- It involves intentions and intuitions.
- It involves rational capacities of a human being.

5) What are the differences between Mill and Bentham on theory Utilitarianism?
- Mill would chose maximum good over individual rights.
- Mill gives preference to individual freedom over maximum utility.
- Mill is more reasonable in choosing the pleasures.
- Mill makes no distinction amongst the different kinds of pleasures.

6) How does the J.S. Mill's theory support the Anti Drug addict rehabilitation programs running across the whole world?
- We should make a qualitative difference in choosing our pleasures.
- We should choose those pleasures which make our life qualitatively better.
- We should go for pleasures only.
- We should rationalize the results or consequences of our pleasures that get through drug-addiction.

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End Term ODD SEMESTER Examination, 2014 - 15

B.Tech. II Year III Sem. (Common to ALL branches)

Subject: - MATHEMATICS - III (AHM – 201)

Time:- 2 Hrs. 30 Minutes **Total Marks:- 40**

Note:

(1) This question paper contains three sections – A, B and C. Section A contains 16 questions, Section B contains 5 questions while Section C contains 4 questions.

(2) Attempt **ALL** questions from Section A, **ANY FOUR** questions from Section B and **ANY THREE** questions from Section C.

(3) Any missing data may be assumed suitably giving proper justification.

SECTION A (16 x 1 = 16)

Instructions: This section has **16** questions. Attempt **ALL** questions of this section.

Q.1. Define Fourier transform of a function F(x).

Q.2. State and Prove linear property of Fourier transform.

Q.3. Solve the integral equation: $\int_0^\infty F(x)\cos px\,dx = \begin{cases} 1-p, 0 \le p \le 1 \\ 0, p > 1 \end{cases}$

Q.4. Find the Z – transform of $\{f(k)\}$ where $f(k) = k(k-1)$ where $k \geq 0$

Q.5. Given that $Z\{u_k\} = \frac{2z^2+3z+4}{(z-3)^2}$, $|z| > 3$. Show that $u_1 = 7$.

Q.6. Find the Z – transform of unit step sequence u (k).

Q.7. Let u(x,y) be the real part of an entire function $f(z) = u(x,y) + i\,v(x,y)$ for $z = x + iy \in C$. If C is the positively oriented boundary of a rectangular region R in R^2 then evaluate $\oint_c \left(\frac{\partial u}{\partial y} dx - \frac{\partial u}{\partial x} dy \right)$

Q.8. If $\vec{\phi}$ and $\vec{\psi}$ are irrotational vectors then prove that

$$\text{div}\,(\vec{\phi} \times \vec{\psi}) = 0$$

Q.9. Find the directional derivative of $f(x, y, z) = x + y + z$ at the point $(\frac{3}{4}, \frac{1}{9}, \frac{5}{7})$ in the direction parallel to y – axis.

Q.10. Prove that: $\{\text{grad } f(r)\} \times \vec{r} = \vec{0}$ where $\vec{r} = x\,\hat{i} + y\,\hat{j} + z\,\hat{k}$.

Q.11. Define analytic function.

Q.12. If $u = x + y + z$, $v = x^2 + y^2 + z^2$ and $w = xy + yz + zx$ then show that grad u, grad v and grad w are coplanar vectors.

Q.13. What are Cauchy – Riemann equations?

Q.14. Prove that $u = x^2 - y^2$ is a harmonic function.

Q.15. Define singularity of a function f (z) with a suitable example.

Q.16. Show that: curl (grad φ) = $\vec{0}$ where φ is a scalar point function.

SECTION B

(4 x 3 = 12)

Instructions: This section has 5 questions. Attempt **ANY FOUR** questions of this section.

Q.1. Find the harmonic conjugate of $u = x^3 - 3xy^2$. Also construct the corresponding analytic function $f(z) = u + iv$ and express it in terms of z.

Q.2. Let $f(z) = u(x, y) + i\,v(x, y)$ be an analytic function of $z = x + i\,y$. Then using calculus of residues, Show that:

$$\int_0^{2\pi} f(e^{i\theta}) \cos\theta \, d\theta = \pi f'(0)$$

Q.3. Expand the function

$$f(z) = \frac{1}{z} - \frac{3}{z+1} + \frac{2}{z-2}$$

in the region $0 < |z+1| < 1$

Q.4. Evaluate: $\oint_c \frac{(z^3+1)}{(z^2-1)} dz$ where c: $|z| = \frac{3}{4}$

Q.5. Let C be the positively oriented circle given by $|z+3i| = 2$ then evaluate:

$$\oint_c \frac{dz}{z^2+9}$$

SECTION C

(3 x 4 = 12)

Instructions: This section has 4 questions. Attempt **ANY THREE** questions of this section.

Q.1. If $f(z) = u + iv$ is an analytic function of $z = x + iy$, find $f(z)$ in terms of z if

$$\alpha u - \beta v = e^x (\alpha \cos y - \beta \sin y), \alpha \neq 0, \beta \neq 0$$

Q.2. Expand the function $f(z) = \frac{1}{(z+1)(z+3)}$ in the following regions:

(a) $1 < |z| < 3$

(b) $|z| > 3$

Q.3. Determine the poles of the function

$$f(z) = \frac{z^2}{(z-2)^2(z+3)}$$

and the residue at each pole. Hence evaluate $\int f(z)dz$ where c is a circle, with centre at origin and of radius 4.

Q.4. Using contour integration, Prove that:

$$\int_0^\infty \frac{dx}{(1+x^2)^2} = \frac{\pi}{4}$$

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End Term Examination, 2014-15

B. Tech. Year: II, Semester: III

CSE 211: Data Structures and Algorithms

Time: 2 ½ Hrs. **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemente with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16)**

I. What is the advantage of using a Circular Queue over a queu

II. Draw expression tree for the expression (a + ((b*(c-e))/f)..

III. Write the recursive function for preorder traversal of BST

IV. Write the difference between cursor and th

V. Explain theta Notation in detail.

VI. If the number of leaf nodes in a strictly
the number of edges in the tree.

VII. Define MST with an example.

VIII. Define Quadratic Probing

IX. Write complexity of quicksort algorithm for all the three cases.

X. Instead of using counting sort to sort digits in the radix sort algorithm, we can use any valid sorting algorithm and radix sort will still sort correctly. Give your answer(T/F) and also justify it.

XI. Give four applications of stack.

XII. Define node, degree, siblings, depth/height, level

XIII. How do we store a graph in Computer's memory? Discuss.

XIV. Define M-way Search Tree.

XV. Give four example of in-place sorting algorithm.

XVI. Compare sequential search with Binary Search.

Section B

Note: Attempt any FOUR Questions **(3x4=12)**

I. Illustrate the operation of RADIX-SORT on the following list of English words:

COW, DOG, SEA, ROW, MOB, BOX, TAB, BAR, EAR, BIG, TEA, NOW

II. Write any sorting algorithm based on divide and conquer approach

III. Write a recursive algorithm to search an element in an array using Binary Search.

IV. Write an algorithm for Insertion Sort. Show steps(passwise) to sort the given array using Insertion Sort: **3, 5, 43, 2, 0, 1, 32, 22, 14, 34**

V. Write an algorithm for Selection Sort. Show steps(passwise) to sort the given array using Selection Sort: **19,23,14,9,5,2,7,4,1**

Section C

Note: Attempt any THREE Questions **(4x3=12)**

I. Write a recursive algorithm for Merge Sort. Apply Merge sort algorithm for sorting following integer array. **1, 12, 8, 2, 25, 6, 14, 28, 17, 7, 52, 16, 48, 68, 3, 2**

II. Given input { **4371,1323,6173,4199,4344,9679,1989**} .Find the h(X) by applying (m=10)

c) Division Method

d) Mid square Method

III. Write algorithm of Heapsort. Sort the following elements using Heap Sort Algorithm.(show all the steps):
6, 5, 3, 1, 8, 7, 2, 4

IV. Write short notes on File Structure.

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End Term Examination, Odd Semester 2014-15
B. TECH., II-YEAR, III-SEMESTER
CSE 212: Computer Organization

Time: 2 ½ Hrs. **Max. Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions. 1x16 = 16 marks

I. In immediate addressing the operand is placed.....

II. The decimal roots of equation $x2-(101)_b+(210)_b=0$ are 7 and 3.The value of b is

III. A computer program that converts an entire program into machine language at one time is called a/an...

IV. A digital computer has a common bus system for 16 registers of 32 bits each and is constructed with multiplexers; find the number of multiplexers in the bus.

V. Assume a computer with 3-address instruction format, the number of bytes required to encode an instruction is (Assume each address require 24 bits and 1 byte is required for operation code)

VI. The value of bias in IEEE754 double precision is...

VII. How many check bits are needed to detect single bit error in 2000 bits data word?

VIII. What is the function of control unit in CPU?

IX. Pipeline improves CPU performance due to....

X. Differentiate Micro operation and Microinstruction.

XI. Consider a case where 4 –segment pipeline with clock cycle time 20 ns in each sub operation to execute 100 tasks. Pipeline system will take how much time to complete the tasks?

XII. Enlist the difference between Main Memory and Control Memory.

XIII. The _____ and _____ are two methods to write in cache.

XIV. Let SP=0000000 in the stack. How many items are there in the stack if Full=1 and Empty=0?

XV. How many address lines are needed to address each memory locations in a 2048 x 4 memory chip?

XVI. The communication bus which is used to fetch the address of an instruction from memory is called the....

Section B

Note: Attempt any Four Questions. 3x4 = 12 marks

I. Enlist and explain locality of references?

II. Discuss memory hierarchy in detail.

III. Explain various cache mapping techniques. A computer system has a 4K word cache organized in block set associative manner with 4 blocks per set, 64 words per block. The main memory contains 65536 blocks. How many bits are there in each of TAG, SET and word FIELDS.

IV. Why I/O interface is required in I/O organization? Write the difference between Memory- Mapped I/O and Isolated I/O.

V. What are the different modes of data transfer? Explain the DMA Controller with a block diagram.

Section C

Note: Attempt any Three Questions. 4x3 = 12 marks

I. Discuss the working principle of IO Processors. Illustrate the CPU-IOP communication with help of flow chart.

II. A virtual memory has an address space 16K words, a memory space has 8K words and block sizes of 2K words. The following reference generated by CPU

4 2 0 1 2 6 1 4 0 1 0 2 3 5 7

Show each page reference change if the replacement algorithm used is

(a) FIFO (b) LRU

III. What do you mean by Interrupt? Explain various types of Interrupt.

IV. A Computer uses RAM chips of 512 x 8 and ROM chips of 2048 x 8. How many RAM chips and ROM chips are needed to provide a RAM memory capacity of 2048 bytes and ROM memory of 2048 bytes? Draw the memory address map for the system and show CPU connection with memory.

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End-Term Examination, Odd Semester 2014-15
Course: B. Tech, Year: II , Semester: III
Discrete Mathematics (CSE 213)

Time: 2.5 Hours **Total Marks: 40**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16 Marks)**

(I) Which of the following sets are equal :
$A = \{x: x^2-4x+3 = 0\}$, $B = \{x: x \in N, x < 3\}$,
$C = \{x: x \in N, x \text{ is odd}, x < 5\}$, $D = \{1, 3, 1\}$

(II) What do you mean by cardinality of sets?

(III) Find the number of relations from $A = \{a, b, c\}$ to $B = \{1, 2\}$.

(IV) If $n(U) = 80$, $n(A) = 27$, $n(B) = 19$ and $n(AUB)' = 31$. Use Venn diagrams to find $n(A \cap B)$.

(V) Consider the following relations on the set $A = \{1, 2, 3, 4\}$ and determine which of the relations are reflexive.
$R1 = \{(1,1), (1,2), (2,3), (1,3), (4,4)\}$,
$R2 = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$, $R3 = A \times A$

(VI) What rules of inference is used in this famous argument? "All men are mortal. Socrates is a man, therefore, Socrates is mortal".

(VII) How many 4-digits numbers can be formed by using the digits 1, 3, 5, 8, 9 when repetition of digits is allowed?

(VIII) A coin is tossed 9 times. What is the probability of getting at least 2 heads?

(IX) A group of three people is selected at random. What is the probability that all three people have different birthdays?

(X) Show that $((P \vee \neg Q) \wedge (\neg P \vee \neg Q)) \vee Q$ is a tautology.

(XI) Let A= {1, 2, 3, 4, 5}. Determine the truth value of each of the following.(a) $(\exists x \in A)(x+3=10)$ (b) $(\forall x \in A)(x+3<10)$

(XII) What do you mean by quantifiers? Explain it with the help of Example.

(XIII) Define a bipartite graph.

(XIV) Show that the set {1, -1, i, -i} where $i=\sqrt{-1}$ is an abelian group with respect to multiplication as a binary operation.

(XV) Show that if (G, *) is a group, then the identity element in G is unique.

(XVI) Let G be (Z, +) i.e., the group of integers under addition and let $f: G \to G$ defined by $\phi(x) = 3x \quad \forall x \in G$. Show that f is homomorphism.

Section- B

Note: Attempt any Four Questions **(3x4=12 Marks)**

(I) Define subgroup of a group. If H,K are subgroups of G, prove that H∩K is also subgroup of G.

(II) Define spanning tree of a graph. Show that a Hamiltonian path is a spanning tree.

(III) Define group. Prove that $(Z_6, \otimes_6)$ is an abelian group of order 6.

(IV) Show that the set of numbers of the form {a+ b√2: a, b ∈R} is an abelian group with respect to addition of real numbers.

(V) Define the edge connectivity and vertex connectivity of a connected graph. Prove that "Every circuit has an even number of edges in common with any cut set."

Section- C

Note: Attempt any Three Questions **(4x3=12 Marks)**

(I) Show that $(Z_6, \oplus_6, \otimes_6)$ is a ring with zero divisors.

(II) Let (G,*) and (G1,*) be two groups and let f:G $\Rightarrow$ G1 be a homomorphism from G to G1.then prove that:

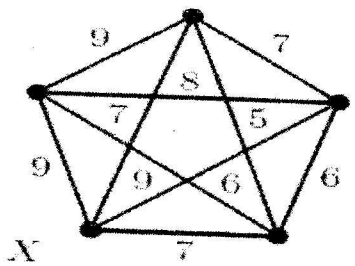
i. f(e)=e1 where e and e1 is the identity in G1.

ii. $f(a^{-1})=(f(a))^{-1}$

iii. Ker f is a normal subgroup of G.

(III) State and prove Lagrange theorem.

(IV) How many Hamiltonian cycles are there in this graph?
Delete the vertex labeled X (and its incident edges).
Find a minimum weight spanning tree on the remaining sub graph. Hence show that the weight (or length) of a solution to the travelling salesman problem is at least 32.



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End Term Examination, Odd Semester 2014-15

B. TECH., II-YEAR, III-SEMESTER

ITE 211:Object Oriented Programming

Instructions:-

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x16=16)**

I. What are the three types of modeling in UML?

II. How do we represent private, public and protected in class diagrams?

III. Elaborate the meaning of degree of association.

IV. Define the term qualified attributes?

V. Name the eight primitive Java data types.

VI. Which object oriented concept is achieved by using overloading and overriding?

VII. When do we use Abstract methods?

VIII. What are the two ways in which a Thread can be created?

IX. Explain the use of parseInt() method using an example.

X. What is the role of init () method under applets?

XI. What is the difference between Swing and AWT components?

XII. What is the default layout for an applet, frame, and panel?

XIII. Why do we use JDBC in java?

XIV. What is the difference between an Applet and a Servlet?

XV. What is the role of stub in RMI?

XVI. What are the different types of EJB?

Section B

Note: Attempt any Four Questions **(3x4=12)**

I. Explain the steps to connect any java application with the database using JDBC.

II. What is a Servlet? Explain the life cycle of a Servlet.

III. Explain the principles of Remote Method Invocation with block diagram.

IV. What is Java Bean? Differentiate between Java Bean and EJB.

V. What are Java Web Services? Describe Composite Computing Model.

Section C

Note: Attempt any Three Questions **(4x3=12)**

I. What are the four types of JDBC drivers in JAVA? Explain with suitable diagrams.

II. Explain getParameter() method used in servlet. What difference does it make if a servlet receive data using get method or post method? Develop a servlet which prints "Hello World" on the browser window.

III. Define Bean Development Kit in JAVA. What are the steps to create a new bean in JAVA?

IV. Develop a client-server application to implement RMI in java.